SCHEME AND SYLLABUS

(W.E.F. ACADEMIC SESSION 2024-25)

FOR

Bachelor of Computer Applications (BCA)

As per UGC Curriculum & Credit Framework for Undergraduate Programme (CCFUP) (Dec 2022)

Offered by

University School of Information, Communication & Technology
GGSIPU at Affiliated Institutions of the University

GURU GOBIND SINGH
INDRAPRASTHA
UNIVERSITY

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Vision of the School

Create High-Quality Engineering and Computer Application Professionals

Mission of the School

To serve humanity by creating professionally competent, socially sensitive engineers with high ethical values who can work as individuals or in groups in multicultural global environments.

Approval History:

1. Scheme of study of BCA programme, its first year detailed syllabus and implementation rules approved by Board of Study of University School of Information, Communication and Technology on 15.12.2023.

2. Scheme of study of BCA programme, its first year detailed syllabus and implementation rules approved by Academic Council on 20.12.2023.

Bachelor of Computer Applications Curriculum Framework as per CCFUP

Aim: In accordance with the revised UGC Curriculum and Credit Framework for Undergraduate Programmes, the Bachelor of Computer Applications (BCA) programme w.e.f academic session 2024-25 curriculum framework shall be as follows:

The BCA programme covers basic to advanced level concepts in Computer Science, Computer Applications and Information Technology. The theoretical concepts in class room teaching and hands on machine experience in digital laboratories using computer program using computers and standard tools widen the horizon of the students. With the acquired knowledge, students can solve the problems in various domains using computers.

Eligibility Criteria: The students are required to refer the eligibility criteria provided in the admission brochure for BCA programme for the specific academic session. As per the Admission Brochure for the Academic Session (AS) 2023-24, the eligibility criteria for BCA programme is as follows:
"Pass in 12th Class of 10+2 of CBSE or equivalent with a minimum of 50% marks in aggregate* with pass in English (core or elective or functional). Mathematics or Computer Science / or other subject related to Computer Science".

OR

"Three-year Diploma in a branch of Engineering from a polytechnic duly approved by All India Council for Technical Education and affiliated to a recognized examining body with a minimum of 50% marks in aggregate".

Duration of the BCA Programme, Provision of Multiple Exit and Awarding Certificate, Diploma, and Degrees

The maximum duration of programme will be as per regulations and recommendations of the statutory bodies governing the BCA programme. As per existing UGC CCFUP, the total duration for completing the BCA programme from the date of admission in first year to completion of Certificate / Diploma / 3 years' degree/ 4 years Hons. degree (including Breaks) shall not exceed 7 years. The detailed qualification award (Certificate / Diploma/ Degree), exit options and other requirement is given in Table1. Grading System shall be as per Ordinance 11 of the University.

Table1: Duration and qualification Award Details:

No of years of completion	Qualification Awarded	Exit Options	Credit Requirement	Entry Option	Remarks
One Year (I & II Semester)	BCA Certificate	Students have an option of exiting the programme after completion of first year (Semester I and semester II)	Students are required to complete Minimum 52 credits including vocational course which is mandatory.	Students who exit with a BCA certificate are permitted to re-enter within three years from the date	Students are required to secure minimum 48 credits during first year and 4 credits in work based vocational courses during summer break after first year. ii. Students may be permitted to take a break from the study during the period of BCA programme but the total duration for completing the BCA

				when student took break and may complete the BCA programme.	programme from the date of admission in first year shall not exceed 7 years.
Two Years (I to IV Semesters)	BCA Diploma	Students have an option of exiting the programme after completion of second year (Semester III & IV)	Students are required to complete Minimum 96 credits and also secure 4 credits in skill based vocational course either offered after first year or after second year during the summer break time.	Students who exit with a BCA diploma are permitted to re-enter within three years from the date when student took break and may complete BCA programme.	i. Students exiting the programme after securing 96 credits in initial two years of admission will be awarded BCA Diploma provided they secure additional 4 credit in skill based vocational courses offered after first year or after second year during the summer break time. ii. Students may be permitted to take a break from the study during the period of BCA programme but the total duration for completing the BCA programme from the date of admission to first year shall not exceed 7 years.
Three Years (I to VI Semesters)	BCA	Students will be allowed to exit after completion of the 3rd year (V & VI semesters)	Students are required to complete Minimum 146 credits and also secure 4 credits in Summer Internship during the summer break time after 2nd year.	Students who exit with a BCA are permitted to re-enter within two years and complete the four year BCA, but the student is not entitled to re-enter for BCA (Honours).	Students who want to undertake 3-year BCA programme will be awarded BCA Degree upon securing 146 credits Students may be permitted to take a break from the study during the period of BCA programme but the total duration for completing the BCA programme from the date of admission year shall not exceed 7 years.
Four Years (I to VIII Semesters)	BCA Honours (Hons.)		Hons. shall only be awarded if the student acquires full credits in every semester with a total CGPA of above or equal to 7.5, and the degree is	--	Students will be awarded BCA (Hons.) Degree

			awarded after the immediate completion of the 4th year from the year of admission. No Hons. shall be conferred if the degree requirements are not completed in the minimum dura tion.		

It is envisioned that the graduates passing out BCA programme, will achieve the following BCA programme specific Learning Outcomes (PLO) and Generic Learning Outcomes (PO) as detailed in Table 2.

Table 2: Programme specific Learning Outcomes (PLO) and Generic Learning Outcomes (GLO)

Programme Specific Learning Outcomes (PLOs)	**Description**
PSO 1	Comprehensive knowledge and coherent understanding of the Computer Applications in various domains and emerging developments associated with the Computer Science and Information Technology
PSO 2	Practical, professional, and procedural knowledge required for carrying out professional or highly skilled work/tasks in the field of computer science and Information Technology, including knowledge required for undertaking self-employment initiatives, and knowledge and mindset required for entrepreneurship involving enterprise creation, improved product development, or a new mode of organization.
PSO 3	Skills in areas related to specialization in the chosen disciplinary/interdisciplinary area(s) of learning including wide-ranging practical skills, involving variable routine and non-routine contexts relating to the Computer Applications
PSO 4	Capacity to extrapolate from what has been learned, translate concepts to real-life situations and apply acquired competencies in new/unfamiliar contexts, rather than merely replicate curriculum content knowledge, to generate solutions to specific problems.
Programme Outcomes (PO)	The student should be able to demonstrate the capability to:
PO1	***Disciplinary Knowledge:*** Apply the knowledge of computer application concepts and domain knowledge to solve the problems in IT domain/IT industry
PO2	***Problem Analysis:*** Identify, formulate, review research literature, and analyse complex computer application problem at their workplace and for the society.
PO3	***Design /Development of Solutions:***

	Design and evaluate solutions for computer applications problems, and design the processes that meet specified needs with appropriate consideration for the public health, safety, cultural, societal, and environmental considerations.
PO4	***Modern Tool Usage:*** Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computer application activities, with an understanding of the limitations.
PO 5	***Professional Ethics:*** Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
PO 6	***Life-long Learning:*** Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.
PO 7	***Project management and finance:*** Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 8	***Communication Efficacy with Cooperation/teamwork*** Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions. Function effectively as an individual and as a member or a leader.
PO 9	***Societal and Environmental Concern:*** Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practices.
PO 10	***Innovation and Entrepreneurship*** Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

Curricular components of the BCA programme

The BCA programme curriculum consists of core courses and interdisciplinary / multidisciplinary courses from other disciplines, ability enhancement courses (language courses), skill enhancement courses, and a set of value added courses. Vocational courses are also added which will help the students to equip with job- oriented skills. The minimum credit requirements for each component for 3-year BCA and 4-year BCA (Hons.) are given in Table 3. The distribution of credits across semesters for BCA programme is given in Table 4.

Table 3: Category Wise Minimum Credit Requirements

S.No.	Broad Category of Course	Minimum Credit Requirement	
		3-year BCA	4-Year BCA (Hons.)
1	Core Courses (CCT/ CCP/PCE)	114	124
2	Multidisciplinary Courses	09	09
3	Ability Enhancement Courses (AEC)	08	08
4	Skill Enhancement Courses (SEC)	08	10
5	Value Added Courses (VA)	08	08
6	Summer Internship (SI)	04	04

7	Major/ Minor Project (PRJ)	06	22
	Total	**157**	**185**

The student is required to complete one Minor Project in the sixth semester, Major Project-1 in seventh semester and Major Project-2 in eight semesters, all from any emerging technologies.

Table 4: Distribution of Credits across Semesters for BCA Programme

S.No.	Broad Category of Course	Semester (Credits)							
		I	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**
1	CCT/ CCP/PCE	20	21	20	16	17	20	10	-
2	Multidisciplinary Courses	-	-	3	3	3	-	-	-
3	Ability Enhancement Courses (AEC)	3	3	-	2	-	-	-	-
4	Skill Enhancement Courses (SEC)	1	-	4	2	-	1	2	-
5	Value Added Courses (VA)	2	2	2	-	-	2	-	-
6	Summer Internship (SI)	-	-	-	-	4	-	-	-
7	Major/ Minor Project (PRJ)	-	-	-	-	-	6	6	10
	Total	**26**	**26**	**29**	**23**	**24**	**29**	**18**	**10**

***Multidisciplinary Generic Elective (GE) for BCA Students**

GE-1 (choose any One)

(i) Principles of Management & Organizational Behaviour

(ii) Any One Paper Offered as open elective by other School /Department / Programme

GE-2 (choose any One)

(i) Digital Marketing

(ii) Principles of Accounting

(iii) Any One Paper Offered as open elective by other School / Department / Programme

GE-3

(i) Introduction to Management & Entrepreneurship Development

(ii) MOOC Course

Vocational Courses Recommendations:

The objective of encouraging the students to undertake a vocational course during summer break after First Year (Second semester) and, also after third year (Sixth semester) for four years programme are

- To ensure the development of capabilities across a range of disciplines including sciences, social

sciences, arts, humanities, languages, as well as vocational subjects, a student can undertake programmes /courses of study relating to Languages, Literature, Music, Philosophy, Art, Dance, Theatre, Statistics, Pure and Applied Sciences, Sports, etc., and other such subjects needed for a multidisciplinary and stimulating learning environment

- To prepare professionals in cutting-edge areas that are fast gaining prominence, such as Artificial Intelligence (AI), 3-D machining, big data analysis, and machine learning, Block chain Technology, Full stack web development, Robotic Process Automation or any other emerging technology with important applications to health, environment, and sustainable living the students are also encouraged to undergo training in the emerging technologies which are not part of their curriculum.
- To find a job for those students who exit before completing the programme.

Summer Internship Recommendations

Students can undertake internships with local industry, businesses etc., or with faculty and researchers at their own or other Higher Educational Institutes (HEIs).
Summer Training (Conducted at the end of the 4th Semester and evaluation to be carried in 5th Semester) Report and Viva – Voce: Students will undergo summer training/industry visit/In-house training/In-house project during the summer break after the completion of 4th semester. This will help the students to engage with all aspects of their learning and facilitate their improvement in the employability. A report of the Summer Internship is required to be submitted to the College/ Institution. Viva-voce examination will be conducted based on the report submitted by the student. A panel of examiner will be appointed by the HOD/ Director of the Institution for internal evaluation out of 40 marks. External evaluation of 60 marks will be conduct by the examination division of the university.

Generic (Open) Electives for other undergraduate programmes

The following Core courses (as mentioned in Table 5) of BCA programme may be offered as Generic Elective for other undergraduate programmes. Maximum number of students from other School / Department / Programme should not exceed 20% of total intake for the programme.

Table 5: Generic (Open) Electives offered by BCA programmes for other undergraduate programmes

S.No.	Semester	Subject Code	Subject Name
1	I	BCA 105T BCA 105P	Web Technologies Web Technologies Lab
2	II	BCA 102T BCA 102P	Database Management System Database Management System Lab
3	III	BCA 201T BCA 201P	Python Programming Python Programming Lab

Major/ Minor Project Recommendations

The student shall undertake Minor Project in 6th Semester, Major Project -1 in 7th Semester and Major Project-2 in 8th Semester, all in any of the emerging areas. Four year students not undertaking Major Project-2 will do an industry internship throughout the eighth semester in lieu of a Major project-2 and will be awarded BCA (Hons.). The students who secure 185 credits, including 22 credits from Minor Project, Major Project-1 and Major Project-2 / internship, shall be awarded BCA (Hons.).

The semester wise evaluation scheme of BCA Programme are mentioned from Table 6 to Table 13.

SEMESTER WISE EVALUATION SCHEME

Table 10: FIFTH SEMESTER

Code No.	Paper		Course Type	L	T/P	Credits	Marks Internal	Marks External	Max Marks
Core Course Theory (CCT)									
BCA 301T	Computer Networks		**CCT**	4	--	4	40	60	100
BCA 303T	Artificial Intelligence		**CCT**	4	--	4	40	60	100
Core Course Practical (CCP)									
BCA 301P	Computer Networks Lab		**CCP**	-	2	1	40	60	100
BCA 303P	Artificial Intelligence Lab		**CCP**	-	4	2	40	60	100
Choose Any One Theory and Respective Practical (Programme Core Electives (PCE))									
BCA 305	BCA 305T	Natural Language Processing	**PCE**	4	-	4	40	60	100
	BCA 305P	Natural Language Processing Lab	**PCE**	-	4	2	40	60	100
BCA 307	BCA 307T	Network Security	**PCE**	4	-	4	40	60	100
	BCA 307P	Network Security Lab	**PCE**	-	4	2	40	60	100
BCA 309	BCA 309T	Full Stack Development using Java	**PCE**	4	-	4	40	60	100
	BCA 309P	Full Stack Development using Java Lab	**PCE**	-	4	2	40	60	100
Summer Internship (SI)*									
BCA 311*	Summer Training*		**SI**	0	0	4	40	60	100
Multidisciplinary (Generic Compulsory) (Choose Any One)									
BCA 313T	Introduction to Management & Entrepreneurship Development		**GE-3**	3	0	3	40	60	100
BCA 315T**	MOOC Course**		**GE-3**	3	0	3	100	-	100
	Total					**24**			**800**

* Summer Training (Conducted at the end of the 4th Semester and evaluation to be carried in 5th Semester) Report and Viva – Voce: Students will undergo summer training/industry visit/In-house training/In-house project during the summer break after the completion of 4th semester. Report of the same is required to be submitted to the College/ Institution. Viva-voce examination will be conducted based on the report submitted by the student. A panel of examiner will be appointed by the HOD/ Director of the Institution for internal evaluation out of 40 marks. External evaluation of 60 marks will be conduct by the examination division of the university.

** NUES (Non – University Examination Subject) – Only Internal Assessment by the Institute)
The students can learn these courses / technologies through self-paced MOOC courses or through regular courses offered by the institute during the summer break on a fast-track mode. The broad guidelines for undertaking MOOC courses are as follows:

1. MOOC Course should be done from SWAYAM/NPTEL as per the guidelines of UGC.
2. For securing the credits, the student is required to complete the assessment of the course and to provide the certificate of the course done from SWAYAM/NPTEL.
3. The fees (if any) for the registration and/or assessment of the MOOC course must be borne by the student only.
4. If the student secures more than 4 credits for the MOOC Course even then 3 credits shall be considered for this subject and the grade/marks provided by assessing authority shall be transferred to the university by the institution where the student is studying. The University's Examination Division shall take the result of the MOOC course on record and the final result must accommodate the results received from such MOOC courses. The student must submit the result of such papers to their respective institution. All the results for the MOOC courses may be submitted before the completion of other requirements including credits requirement.

FIFTH SEMESTER SYLLABUS

Course Code: BCA 301T **L T C**
Course Name: Computer Networks **4 0 4**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following: -

1. To study different types of media, multiplexing, switched networks, the Internet, TCP/IP suite, fiber-optic communications and the state-of-art networking applications.
2. To develop an understanding of different components of computer networks, various protocols, modern technologies and their applications.
3. Identify and discuss the underlying concepts of IPv4 & IPv6 protocols, along with their characteristics and functionality.
4. Details of IP operations in the Internet and associated routing principles
5. Analyzing various layering protocols in computer networks.

PRE-REQUISITES:

1. None

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO#	Detailed Statement of the CO
CO1	Utilize the fundamentals of data communication and networking to identify the topologies and connecting devices of networks.
CO2	Understand and describe the layered protocol model (OSI and TCP/IP model)
CO3	Evaluate and implement routing algorithms and Router basic configuration.
CO4	Evaluate the protocols and Principles in computer networking

Course Outcomes	Program Outcomes (Scale-1: very low, 2: low, 3: medium, 4:high)									
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**
CO1	4	2	2	4	-	3	-	-	-	1
CO2	4	4	2	4	-	2	4	-	-	-
CO3	4	4	1	4	-	3	2	-	-	-
CO4	4	4	-	4	2	3	1	-	-	-

UNIT – I
No. of Hours: 10; TB1[Chapter-1], TB2[Chapter-1,2]

Basic Concepts: Components of data communication, distributed processing, Line configuration, topology, transmission mode, and categories of networks. **OSI and TCP/IP Models:** Layers and their functions, comparison of models. **Transmission Media:** Guided and unguided, Attenuation, distortion, noise, signal to noise ratio (S/N), throughput, propagation speed and time, wavelength, Shannon Capacity.

UNIT – II
No. of Hours: 12; TB1[Chapter-2,3], TB2[Chapter-3,9]

Telephony: Multiplexing, WDM, TDM, FDM, circuit switching, packet switching and message switching. **Data Link Layer:** Types of errors, Framing (character and bit stuffing), error detection & correction methods (LRC,VRC,CRC,2D Parity, CHECKSUM, HAMMING CODE); Flow control; Protocols: Stop & wait ARQ, Go-Back- NARQ, Selective repeat ARQ.

UNIT – III
No. of Hours: 12; TB1[Chapter-5], TB2[Chapter-18,19,20,22]

Network Layer: Internetworking & Devices: Repeaters, Hubs, Bridges, Switches, Router, Gateway, Modems; Addressing: IPv4 and IPv6 addressing, IPv4 subnetting;
Routing: Introduction to Routing, Routing Algorithms (Adaptive, Non-Adaptive), Types of Routing Tables, Flooding, Shortest Path Routing, Distance Vector Routing, Link State Routing

UNIT – IV
No. of Hours: 10; TB1[Chapter-6,7], TB2[Chapter-23, 24,25]

Transport and upper layers in OSI Model: Transport layer functions and Protocols, connection management, functions of session layers, Presentation layer, and Application layer.

TEXT BOOKS:
TB1. A. S. Tanenbaum, "Computer Networks", Pearson, 5th Edition, 2013.
TB2. Behrouz A. Forouzan, "Data Communication and Networking", TMH, 5th Edition, 2017

REFERENCE BOOKS:
RB1. Douglas E. Comer, "Computer Networks and Internet", Pearson, 6th Edition, 2018
RB2. W. Stallings, "Data and Computer Communication", Macmillan Press, 8th Edition, 2016
RB3. Leinwand, A., Pinsky, B. (2020). Cisco router configuration. United Kingdom: Cisco Press.

Course Code: BCA 303T **L T C**

Course Name: Artificial Intelligence **4 0 4**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To learn the basics of designing intelligent agents that can solve general purpose problems.
2. To represent and process knowledge, plan and act, reason under uncertainty and can learn from experiences.

PRE-REQUISITES: Basic Programming Skills

COURSE OUTCOMES(COs): After completion of this course, the learners will be able to: -

CO#	Detailed Statement of the CO
CO1	To understand elements constituting problems and learn to solve it by various uninformed and informed (heuristicsbased)
CO2	To understand formal methods for representing the knowledge and the process of inference to derive new representations of the knowledge.
CO3	Analyze and apply the notion of uncertainty and some of probabilistic reasoning methods to deduce inferencesunder uncertainty
CO4	Apply some mechanisms to create and improve AIsystem.

Course Outcome	Program Outcomes (Scale-1:very low, 2: low, 3 medium, 4: high									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	1	2	-	3	-	-	-	-	-	-
CO2	2	2	-	3	-	-	-	-	-	-
CO3	2	1	-	4	-	-	-	-	-	-
CO4	-	-	-	2	-	2	-	3	-	-

UNIT–I

No. of Hours: 12 Chapter/Book Reference: TB1 [Chapters - 1, 2, 3]; TB2 [Chapters- 1, 3, 4]

Overview of AI: Introduction to AI, Importance of AI, AI and its related field, AI techniques, Problem solving agents, Criteria for success.

Problems, problem space and search: Defining the problem as a state space search, Depth First Search, Breadth First search, Production Systems and its characteristics, Issues in the design of the search programs.

Heuristic search techniques: Generate and test, hill climbing, best first search technique, A*, AO*, problem reduction, constraint satisfaction.

UNIT–II

No. of Hours: 12 Chapter/Book Reference: TB1 [Chapters - 5, 6]; TB2 [Chapters - 7, 8, 9, 10] RB1 [Chapters - 5, 6, 7]

Knowledge Representation: Definition and importance of knowledge, Knowledge representation, various approaches used in knowledge representation, Issues in knowledge representation, Semantic net frame.

Logical Reasoning: Logical agents, propositional logic, inferences, Syntax and semantics of First Order Logic, Inference in First Order Logic Knowledge Base, forward chaining, backward chaining, unification, resolution

UNIT–III

No. of Hours: 10 Chapter/Book Reference: TB1 [Chapters - 7, 8, 15]; TB2 [Chapters - 13, 14]

Handling Uncertainty: Non-Monotonic Reasoning, Probabilistic reasoning, Bayes 'Theorem, Certainty factors and Rule-based Systems, Bayesian Networks, Dempster-Shafer Theory, Introduction to Fuzzy logic. Fuzzy set definition & types. Membership functions. Designing a fuzzy set for a given application

Natural Language Processing: Introduction, Syntactic Processing, Semantic Processing, Pragmatic Processing.

UNIT–IV

No. of Hours: 10 Chapter/Book Reference: TB1 [Chapter 17]; TB2 [Chapters - 18, 19]

Learning: Introduction to Learning, Rote Learning, learning by taking advice, learning in problem solving, learning from examples: Induction, Explanation-based Learning, Discovery, Analogy, Neural Networks, and Genetic Learning.

Expert System: Introduction to expert System, Case study of Expert system (Prolog/any programming Language).

TEXT BOOKS:

TB1. Rich and Knight, "Artificial Intelligence", Tata McGraw Hill, 1992.

TB2. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach",Prentice Hall, Second Edition (Indian reprint: Pearson Education)

REFERENCE BOOKS:

RB1. George F.Luger Artificial Intelligence Pearson Education

RB2. Ben Coppin Artificial Intelligence Illuminated Jones and Bartlett Publisher

Course Code: BCA 305T **L T C**
Course Name: Natural Language Processing **4 0 4**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to Natural Language Processing and their applications.

PRE-REQUISITES:

Python programming language

COURSE OUTCOMES(COs):

After completion of this course, the learners will be able to:-

CO#	Detailed Statement of the CO
CO1	Understand NLP fundamentals, Python-based text processing, and its applications.
CO2	Apply text preprocessing, corpora access, representation methods
CO3	Exploring language modelling techniques
CO4	Acquire skills in tagging, classification, sentiment analysis, and apply to real-world challenges.

Course Outcomes	Program Outcomes (Scale – 1: very low, 2: low, 3: medium, 4: high)									
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**
CO1	3	1	-	4	-	2	-	-	-	2
CO2	2	2	2	3	-	1	1	-	-	2
CO3	3	3	3	2	-	2	2	-	-	3
CO4	3	2	1	2	2	2	2	-	-	3

UNIT – I
No. of Hours: 10 Chapter / Book Reference: TB1 [Chapters 1, 5], TB2 [Chapters 1]

Introduction to NLP: Overview of Natural Language Processing, History and evolution of NLP, Applications of NLP in real-world scenarios, Approaches to NLP
Computing with Language: Texts and Words, A Closer Look at Python: Texts as Lists of Words, dictionaries, Ethical considerations and bias in NLP, Current research trends and emerging applications in NLP

UNIT – II
No. of Hours: 10 Chapter / Book Reference: TB1[Chapters 2, 3], TB2[Chapters 2, 6]

Introduction to corpora and text data: Accessing Text Corpora and Lexical Resources, Conditional Frequency, Lexical Resources, WordNet, NLP Pipeline, Strings: Text Processing at the Lowest Level, Text preprocessing techniques: tokenization, stemming, lemmatization

UNIT – III

No. of Hours: 10 Chapter / Book Reference: TB1 [Chapters 5], TB2 [Chapters 3, 6]

Language Modeling, probability theory and statistical language models, Syntactic analysis and parsing techniques, Vector space models, One-hot encoding, Bag-of-Words (BoW) model, TF-IDF (Term Frequency-Inverse Document Frequency) representation, n-gram models, training and test sets, evaluating and smoothing

UNIT – IV
No. of Hours: 10 Chapter / Book Reference: TB1 [Chapters 6, 7], TB2 [Chapters 7]

NLP Tasks and Techniques: Part-of-speech tagging, Named Entity Recognition (NER), Sentiment analysis, Supervised Text Classification, Evaluation, Naïve Bayes Classifiers, Chatbot: Dialog system pipeline, components
Case study: Students will work on a real-world NLP problem, applying techniques learned throughout the course and presenting their findings.

TEXT BOOKS:

TB1. Steven Bird, Ewan Klein, Edward Loper, “Natural Language Processing with Python”, O'Reilly Media, Inc., 2021
TB2. Jurafsky & Martin, "Speech and Language Processing”, Pearson Publication, 2nd Edition, 2013.
TB3. Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, “Practical Natural Language Processing”, O'Reilly Media, Inc. 2020
TB4. Tanveer Siddiqui, U. S. Tiwary, "Natural Language Processing and Information Retrieval", Oxford University Press, 2008

REFERENCE BOOKS:

RB1. Delip Rao, Brian McMahan, “Natural Language Processing with PyTorch”, O'Reilly Media, Inc., 2019
RB2. Jacob Einsein, “Introduction to Natural Language Processing”, MIT Press, 2nd Edition, 2019.

Course Code: BCA 307T **L T C**

Course Name: Network Security **4 0 4**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

1. Students will be able to learn about basic security issues and concepts of network security.
2. To understand the design principles of firewalls and different types of firewalls.
3. To understand the concept of authentication applications and protocols.
4. To understand different network security algorithm by using Program.
5. Students will be able to understand the computer network and secure network communication issues along with their remedies.

PRE-REQUISITES:

1. Computer Networks
2. Programming Languages (C, C++, Java)

COURSE OUTCOMES(COS):

After completion of this course, the learners will be able

CO#	Detailed Statement of the CO
CO1	Define and explain the issues and basic concepts of Network Security. To understand different network models.
CO2	To Explain, understand and summarize the concepts, types and features of Firewall. Firewall by using some suitable examples.
CO3	Explain and implement working of authentication, authorization, Email Security.
CO4	Classify and organize the architecture of network security management.
CO5	Evaluate different Network Security algorithms with the help of program.

Course Outcome	Program Outcomes (Scale-1:very low, 2: low, 3 medium, 4: high									
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**
CO1	2	1	1	2	1	1	1	1	1	1
CO2	3	2	2	2	3	1	2	1	1	1
CO3	2	2	1	1	2	1	2	2	1	1
CO4	2	1	1	2	2	1	3	2	2	2
CO5	2	3	1	2	1	1	1	2	2	2

UNIT-1
No. of Hours: 12 Chapter/Book Reference: TB1 [Chapters-1, 2, 3] and TB2 [Chapters - 1, 2]

Introduction to Network Security and related issues- authentication, confidentiality, integrity, anonymity, etc. Network Security- Threats, Attacks and Vulnerabilities, Network security services and mechanisms. Network Security Models, Public Key Infrastructure

Cryptography: Introduction to Cryptography, Symmetric Encryption, Asymmetric Encryption, Hash Function, Digital Signature, Message Authentication, Digital Signature Algorithm, Cryptography protocols- SSL/TLS (Secure socket layer/ Transport layer Security), IP Sec (IP Security), VPN (Virtual Private Network), Key Management. Quantum Cryptography

UNIT-2
No. of Hours: 11 Chapter/Book Reference: TB1 [Chapter-5]

Firewalls: Overview, Firewall Architecture, Features of Firewalls, Design Principles of Firewalls, Types- Packet Filtering, Application Gateway, Proxy Firewalls, Firewall Rule Configuration, How Firewall support for VPN technologies, Logging and Monitoring.

UNIT-3
No. of Hours: 11 Chapter/Book Reference: TB1 [Chapters-3, 4, 5]

Authentication Applications & Authentication Protocols, E-Mail security, Pretty Good Privacy (PGP), Secure Multipurpose Internet Mail Extensions (S/MIME), Encapsulating security payload (ESP).

Network Management Security: Overview of SNMP Architecture, Available software platforms/case tools (Wireshark/Solar Winds/ KF Sensor/Kali Linux/ NMAP), Configuration Management.

UNIT-4
No. of Hours: 10 Chapter/Book Reference: TB2 [Chapters - 8] and TB3 [Chapters-4, 5]

Intrusion Detection and Prevention Systems (IDPS), Intruders, Intrusion Detection & Prevention System, Types of IDPS- Host, Network and Hybrid. Detection Techniques- Signature based Detection, Anomaly-based

Detection, Behavior-based Detection, and Heuristic Detection. Prevention Techniques- Blocking, Alerting, Quarantine, Threat Intelligence Integration.

TEXT BOOKS:

TB1 Kaufman et al., Network Security, Second Edition, Publisher: Prentice Hall, 2002
TB2. Stallings and Brown, Computer Security: Principles and Practice, Fourth Edition, Publisher: Pearson, 2018.
TB3. Practical Network Security: Detecting and Mitigating Network Attacks" by Kim Crawley and Phillip Wylie, 2020

REFERENCE BOOKS:

RB1. Trappe and Washington, Introduction to Cryptography with Coding Theory, Third Edition, Publisher: Pearson, 2020.
RB2. Principles of Information Security: Michael E. Whitman, Herbert J. Mattord, CENGAGE Learning, 4th Edition.
RB3. W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education, 2003

Course Code: BCA 309T **L T C**
Course Name: Full Stack Development using Java **4 0 4**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. The core concepts of both the frontend and backend programming.
2. The latest web development technologies.
3. Managing data using MongoDB.
4. Complete web application development process.

PRE-REQUISITE:

1. HTML, CSS
2. JavaScript

COURSE OUTCOMES:

CO#	Detailed Statement of CO
CO1	Comprehend the basics of JavaScript and TypeScript
CO2	Apply the concepts of Servlets and JSP
CO3	Examine Eclipse and Apply Angular for frontend design
CO4	Create Full Stack Applications using Java and MongoDB

Course Outcomes	Program Outcomes (Scale - 1: Very Low, 2: Low, 3: Medium, 4: High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	4	4	4	4	-	-	-	-	-	3
CO2	4	4	4	4	-	-	-	-	-	3
CO3	4	4	4	4	-	-	-	-	-	3
CO4	4	4	4	4	-	-	-	-	-	3

UNIT I
No. of Hours: 12 Chapter/Book Reference: TB3 [Chapters: 2]

Introduction to Full Stack Development, its role in modern software development and real-life applications. Overview of Typescript and Introduction to ES6, Variable declaration, Arrow functions, Objects & Arrays, Modules, Functions, Classes, Asynchronous Programming.

JSON: Introduction, Data Types, Objects, Schema. Introduction to REST API, WRML.

UNIT II
No. of Hours: 12 Chapter/Book Reference: TB2 [Chapters: 2,4,7-10]

Servlets: Introduction to Java Servlets, Need of Servlets, Servlet Features, Life cycle, Servlets Architecture, Servlets APIs, Form Data, Client HTTP Request, Server HTTP Response, Servlets Exception Handling, Servlets Packages, Interfaces and classes in Javax. Database Access with Servlets

JSP (Java Server Pages): Introduction to JSP, Features of JSP, JSP Architecture of JSP, Working of JSP Engine, Life cycle of JSP, JSP Tags, Static and Dynamic Contents, Directives, Actions, Implicit Objects of JSP, JSP Client Request, JSP Server Response, Handling Exception in JSP, JSTL Tags, JSP API, Difference between Servlet and JSP.

UNIT III
No. of Hours: 12 Chapter/Book Reference: TB1 [Chapters: 20-26]

Introduction to Eclipse and Angular. Angular Life Cyle, Angular Decorators, Angular Component, Angular Module, Angular Data Binding (One way and Two way Data Binding), Angular Interpolation, Angular Property Binding, Angular Event Binding, Angular Data Binding - Two way data binding (ngModel), What are Angular Directives, Angular Structural Directives, Angular nglf Directive, Angular ngStyle Directive, Angular Pipes, Angular Forms, Angular Template Driven Form, Angular Reactive Form

UNIT IV
No. of Hours: 12 Chapter/Book Reference: TB1 [Chapters: 11-15]

Difference between Relational Database and NoSQL, Introduction to Database Connectivity, MongoDB and its datatypes, connecting to MongoDB, managing and manipulating collections, documents and fields.

TEXT BOOKS:

TB1. D. Brad, B. Dayley and C. Dayley, "Node. js, MongoDB and Angular Web Development: The definitive guide to using the MEAN stack to build web applications", Addison-Wesley Professional, 2nd Edition, 2017.

TB2. B. Basham, K. Sierra and B. Bates, "Head first servlets and JSP: passing the Sun certified web component developer exam", O'Reilly Media, Inc., 2nd Edition, 2008.

TB3. C.J. Ihrig and A Bretz,"Full Stack JavaScript Development with MEAN", SitePoint, 1st Edition, 2015.

REFERNCE BOOKS:

RB1. M. Masse, "Rest API – Design Rulebook", O'Reilly, 1st Edition, 2011

Course Code: BCA 313T **L T C**
Course Name: Introduction to Management & Entrepreneurship Development **3 0 3**

INSTRUCTIONS TO PAPER SETTERS:

- Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
- Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
- Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, using Bloom's Taxonomy (BT), in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following: -

1. To inspire students and help them imbibe an entrepreneurial mind-set.
2. To Understand the ecological factors needed to become a successful Entrepreneur
3. To learn to manage finances, GST and Human resource of Start Up effectively
4. To develop an entrepreneurial mind-set engaging in CSR activities

PRE-REQUISITES:

1. None

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to: -

CO #	Detailed Statement of the CO
CO1	To comprehend Entrepreneurship in terms of its theories
CO2	To understand Venture promotion and fund raising for Entrepreneur and examine AI and DS Tools
CO3	To examine the concept of MSME, Finance managing and review successful Women Entrepreneurship case studies
CO4	To grasp the importance of HRM, GST and CSR in terms of Entrepreneurship

Course Outcomes	Program Outcomes (Scale-1: very low, 2: low, 3: medium, 4:high)									
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**
CO1	1	4	4	-	3	3	2	-	-	3
CO2	4	4	-	3	-	3	2	1	-	4
CO3	1	-	3	-	4	4	3	2	-	4
CO4	2	-	3	-	-	4	4	3	3	4

UNIT – I
No. of Hours: 10; TB1[Chapter-1, 3]

Basic Concepts: Introduction to Entrepreneurship Meaning and characteristics of entrepreneurship, Emergence of Entrepreneurial Class. **Theories of Entrepreneurship:** Sociological Theories – Max Weber, F Young and EE Hagen. Economic Theory – Theory of Innovation, Harvard School Theory, Psychological Theories – Theory of Personal Resourcefulness and Theory of Entrepreneurship

UNIT – II
No. of Hours: 12; TB1[Chapter-4,5]

Promotion of Venture: Opportunity Analysis, Environment and Entrepreneurship, Technological Environment, Competitive Factors, Small Scale Industrial Undertakings, Micro, Small and medium Enterprises, steps in setting up of a small-scale Industrial Enterprise, Legal Requirements and Important Acts. **Raising of Funds –** Types, Sources, Capital Structures, Capitalization, venture Capital: Concept, Steps and Sources. **Tools** - AI and Data Sciences Tools to aide Entrepreneurship

UNIT – III
No. of Hours: 10; TB2[Chapter-4, 6, 7], TB3[Chapter-9]

MSME: Introduction – Business Entities – Meaning, Concepts and Nature, Sole Proprietorship, Family Business, MSME's – Definition, Classification, Benefits, Challenges, Procedure to Apply and Government Measures.
Women Entrepreneurship – Kiran Mazumdar Shaw, Dr. Swati Piramal,
Managing Finance – Objectives, Operating and Cost Budget, Cash Budget, Short Term Budget, Long Term Budget and importance of owing IPR

UNIT – IV
No. of Hours: 12; TB1[Chapter-8]

Human Resource Planning: Significance and Steps in HRP, Contract Management, Marketing Methods including Digital Marketing, Social Media Marketing, Case Studies of Successful Startups
GST: GST Definition and Significance, Types of Records maintained under GST, Types of Accounts maintained under GST, Features of GST, Methods of GST Number Registration, Consequences of not maintaining proper GST record
Sustainability and Corporate Social Responsibility (CSR) – Introduction to CSR, Why Firms Engage in CSR? Theories, Institutions and Models of Capitalism.

TEXT BOOKS
TB1. Sangram Keshari Mohanty, "Fundamentals of Entrepreneurship", PHI, 2017
TB2. Dr. Abha Mathur, "Entrepreneurship Development", Taxmann's. July 2021, ISBN: 9789391596088
TB3. T.N. Chabbra, Environmental Development, Sun India, 2021,

REFERENCES:
RB1. Dan Sitarz, "Sole Proprietorship: Small Business Start-up Kit", Nova Publishing Company 2011
RB2. Sabri Boubaker, Duc Khuong Nguyen, "Corporate Social Responsibility, Ethics and Sustainable Prosperity", World Scientific Publishing Co. Pte. Ltd
RB3. Philip Kotler and Nancy Lee, "Corporate Social Responsibility: Doing the Most Good for Your Company and Your Cause", Wiley, 2005

Course Code: BCA 301P	L	T/P	C
Course Name: Computer Networks Lab	0	2	1

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to:

1. To study different types of media, multiplexing, switched networks, the Internet, TCP/IP suite, fiber-optic communications and the state-of-art networking applications.
2. To develop an understanding of different components of computer networks, various protocols, modern technologies and their applications.
3. Identify and discuss the underlying concepts of IPv4 & IPv6 protocols, along with their characteristics and functionality.
4. Details of IP operations in the Internet and associated routing principles.
5. Analyzing various layering protocols in computer networks.

PRE-REQUISITES:

1. None

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:

CO #	Detailed Statement of the CO
CO1	Utilize the fundamentals of data communication and networking to identify the topologies and connecting devices of networks.
CO2	Understand and describe the layered protocol model (OSI and TCP/IP model)
CO3	Evaluate and implement routing algorithms and Router basic configuration.
CO4	Evaluate the protocols and Principles in computer networking

List of Practicals

S. No.	Detailed Statement	Mapping to CO #
	Core Practicals	
1	Introduction to packet tracer simulation tools	CO1
2	Study of different types of Network cables and practically implement the cross-wired cable and straight cable	CO1
3	Study of Network Devices in Detail.	CO1, CO2
4	Study of network IP.	CO1, CO2, CO4
5	Creating a local area network and testing the connectivity within the network	CO1, CO2,CO4
6	Study of basic network command and Network configuration commands	CO1, CO2,CO3,CO4
7	Creating and configuring a simple peer-to-peer network having two PC's and testing the connectivity between them.	CO1, CO2, CO3
8	Interconnecting two different LAN's and testing the connectivity between them.	CO1, CO2, CO3
9	Router Configuration Command line Interface	CO1, CO2, CO3, CO4
10	Implementing Routing Protocol RIP, OSPF, BGP.	CO1, CO2, CO3, CO4

Course Code: BCA 303P	L	T/P	C
Course Name: Artificial Intelligence Lab	0	4	2

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To learn the basics of designing intelligent agents that can solve general purpose problems.
2. To represent and process knowledge, plan and act, reason under uncertainty and can learn from experiences.

PRE-REQUISITES:

1. Basic Programming Skills

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:

CO #	Detailed Statement of the CO
CO1	To understand elements constituting problems and learn to solve it by various uninformed and informed (heuristicsbased)
CO2	To understand formal methods for representing the knowledge and the process of inference to derive new representations of the knowledge.
CO3	Analyze and apply the notion of uncertainty and some of probabilistic reasoning methods to deduce inferencesunder uncertainty
CO4	Apply some mechanisms to create and improve AI system.

List of Practicals

S. No.	Detailed Statement	Mapping to CO #
	Core Practicals	
1.	Write a program to implement Breadth First and Depth First Search	CO1
2.	Write a Program for the Best First Search and A* search algorithm	CO1
3.	Write a program to implement Water Jug Problem	CO1
4.	Write a program to implement 4-Queen Problem	CO1
5.	Write a program to implement AO* algorithm	CO1
6.	Write a program to implement hill climbing & steepest ascent hill climbingalgorithm	CO1
7.	Write a program to implement Travelling Salesman Problem	CO1
8.	(a) Write a program to implement List operations (Nested List, Length, Concatenation, Membership, Iteration, Indexing and Slicing)? (b) Write a programto implement List methods (Add, Append, and Extend & Delete)	CO2
9.	Write a program to implement First Order Predicate using: a. Backward Chaining Forward Chaining	CO2
10.	(a) Write a program to remove stop words for a given passage from a text file usingNatural Language Toolkit (NLTK)?	CO3

	(b) Write a program to implement stemming for a given sentence using NLTK? (c) Write a program to POS (Parts of Speech) tagging for the given sentence using NLTK? (d) Write a program to implement Lemmatization using NLTK?	
11.	Write a program for Text Classification for the given sentence using NLTK?	CO3
12.	Write a program to implement Artificial Neural Network (ANN) for Classification using a dataset	CO4
13.	Write a program to implement Genetic Algorithm for different types of gene representation	CO4
14.	Write a program to implement ANN for Bayesian networks	CO4
15.	Write a program to implement back propagation algorithm using ANN	CO4

Note:

1. In total 10 practicals to be implemented. 2 additional practicals may be given by the courseinstructor.
2. This is a suggestive list of programs. However, the instructor may add programs as per therequirement of the course.

Course Code: BCA 305P	L	T/P	C
Course Name: Natural Language Processing Lab	**0**	**4**	**2**

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. Text preprocessing
2. Text analysis
3. Web scrapping

PRE-REQUISITES:

1. Python programming language

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:

CO #	Detailed Statement of the CO
CO1	Apply lemmatization primitives using python.
CO2	Analyze Lexical analysis on various text corpuses.
CO3	Assess the text classification algorithms on text and speech tagging.
CO4	Create an NLP model for analyzing the text documents.

List of Practicals

S. No.	Detailed Statement	Mapping to CO #
	Core Practicals	
1	Use Python to tokenize a corpus of text and calculate the frequency of words.	CO1
2	Implement stemming and lemmatization algorithms to reduce words to their root forms.	CO1
3	Write Python code to access and load text corpora from different sources such as NLTK.	CO1, CO2
4	Develop a POS tagging system to label words with their respective parts of speech.	CO1, CO2, CO3
5	Create a Named Entity Recognition (NER) system to identify and classify named entities in a text corpus.	CO1, CO2, CO3, CO4
6	Build a sentiment analysis model to classify text documents as positive, negative, or neutral based on sentiment polarity.	CO1, CO2, CO3, CO4
7	Implement a text classification system to categorize text documents into predefined classes (3 or more).	CO1, CO2, CO3, CO4
8	Construct vector representations of text documents using techniques like Bag-of-Words (BoW) model or TF-IDF.	CO1, CO2, CO3, CO4
9	Use probability theory to build language models and evaluate their performance on test sets.	CO1, CO2, CO3, CO4
10	Discuss ethical considerations and biases in NLP systems, and analyze the impact of biased datasets on model performance.	CO1, CO2, CO3, CO4

Course Code: BCA 307P	L	T/P	C
Course Name: Network Security Lab	0	4	2

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. Students will be able to learn about basic security issues and concepts of network security.
2. To understand the design principles of firewalls and different types of firewalls.
3. To understand the concept of authentication applications and protocols.
4. To understand different network security algorithm by using Program.
5. Students will be able to understand the computer network and secure network communication issues along with their remedies.

PRE-REQUISITES:

1. Computer Networks
2. Programming Languages (C, C++, Java)

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:

CO #	Detailed Statement of the CO
CO1	Define and explain the issues and basic concepts of Network Security. To understand different network models.
CO2	To Explain, understand and summarize the concepts, types and features of Firewall. Firewall by using some suitable examples.
CO3	Explain and implement working of authentication, authorization, Email Security.
CO4	Classify and organize the architecture of network security management.
CO5	Evaluate different Network Security algorithms with the help of program.

List of Practicals

S. No.	Detailed Statement	Mapping to CO #
	Core Practicals	
1	Implement Security Monitoring Tools	CO1
2	To study and implement Symmetric Encryption cryptographic systems	CO1, CO5
3	To study and implement Asymmetric Encryption cryptographic systems	CO1, CO5
4	Implement Digital Signatures.	CO1, CO5
5	Demonstrate any intrusion detection and prevention system.	CO4, CO5
6	Demonstrate various NMAP commands	CO4, CO5
7	Demonstrate Network Monitoring & Packet analysis	CO5
8	Demonstrate Port Scanning using KF Sensor	CO5
9	Configuring Windows Firewall and implement user management	CO2, CO5
10	Implement network firewall security	CO1, CO4, CO5

11	Implement Security Monitoring Tools	CO1, CO4, CO5

Note:
1. In total 11 practicals' to be implemented. Two additional practical may be given by the course instructor.
2. This is a suggestive list of programs. However, the instructor may add programs as per the requirement of the course.

Course Code: BCA 309P	L	T/P	C
Course Name: Full Stack Development using Java Lab	0	4	2

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

6. Basic understanding of Object-Oriented Programming Concepts and create classes.
7. Learn building GUI applications using various concepts like JSP, Hibernate, Servlets etc.
8. Working expertise of Frontend, server and Backend design and deployment for dynamic web applications.

PRE-REQUISITES:

1. HTML, CSS
2. JavaScript
3. Basic knowledge of web applications

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:

CO #	Detailed Statement of the CO
CO1	To understand and apply the concepts of Java script, CSS, HTML
CO2	To understand and apply the concepts of GUI programming
CO3	Create full stack applications using Angular.js
CO4	Develop applications using Database connectivity

List of Practicals

S. No.	Detailed Statement	Mapping to CO #
	Core Practicals	
1	Write a program to create a website using HTML, CSS and Javascript.	CO1
2	Write a program to create a simple calculator Application using JS, HTML and JSP.	CO1
3	Create a Simple Login form using JS and Java.	CO1
4	Create a web application to manage the TO-DO list of users where user can login and manage their to-do items.	CO2
5	Write a program to build a Chat module using HTML CSS and JavaScript and Java.	CO1
6	Develop a classifies web application to buy and sell used products using JS and Java.	CO1, CO2

7	Create a simple servlet that generates the plain text.	CO2
8	Develop a leave management system for an organization where users can apply different types of leaves such as casual leave and medical leave. They also view the available number of days.	CO2, CO3
9	Write a program to convert the given number of eggs to number of gross, no of dozens and extras HINT 144 eggs=1 gross 12 eggs=1 dozen Example: no of eggs=13429 Gross 3 dozen 10 extras	CO1
10	Fetch data from an API endpoint using the “fetch” function or a third-party library like “axios.” Display the retrieved data in a component called “UserList” that renders a list of user names.	CO3
11	Create an Angular component called "Head" that displays a title and a subtitle. Use this component in your main AppComponent and pass the title and subtitle as inputs.	CO3
12	Create a Reactive Angular form component called "LoginForm" that includes input fields for username and password. Handle form submission and display an alert message with the entered username and password when the form is submitted.	CO3
13	Create a program that updates customer information inside a database.	CO4
14	Design a full stack application using Angular for frontend that allows student to enter his/her details which will be stored in NoSQL database. Additionally, the front-end should fetch and display a list of all stored users from the server.	CO3, CO4

Course Code: BCA 311* **L T C**
Course Name: Summer Training* **0 0 4**

BCA SUMMER TRAINING GUIDELINES

Objective:

All the students enrolled for **BCA** Programme, have to undergo compulsory summer training/ Project of minimum 06 weeks in an organization in the IT field. The aim of the project is to give the students an integrated experience in solving a real-life problem by applying knowledge and skills gained on completion of theory papers and in-house practical papers during BCA course. It provides an occasion for students to realize the importance of resource and time management, ownership of task towards deliverables, innovation and efficiency in the task management. It also provides a good opportunity for students to build, enhance and sustain high levels of professional conduct and performance and evolves a problem solver frame of mind in students at early stage. It also prepares students for taking up responsible assignments in the corporate establishment.

General Guidelines:

1. The project should be original, of real-life value, and not copied from existing material from any source. A student should ensure that he understands what is expected by preparing a requirement document of his understanding and get it reviewed by the guide.
2. Design document should also be reviewed and code should also be peer reviewed.
3. A user manual has to be prepared and reviewed.
4. Testing has to be thorough and at various levels, followed by an acceptance test based on the requirement document and user manual.
5. Students should follow the steps as discussed in Software Development Life Cycle while writing dissertation and use Software Engineering Methodologies for development of deliverables, mere programming will not be sufficient. Students must note that interviewers for job are often more interested in the problem solved, alternatives that could have been tried and the benefits derived from the developed application, rather than just implementation details.
6. **One Project will be submitted only by one student.** However, if the nature of the project is very big & large enough to be divided in different independent big modules having an estimated required effort of minimum 06 weeks to be developed by one person, can be taken up and designed in such a way that every student will be responsible for one module and will submit only that specific module as it were a complete software project.
7. Every student has to get his / her synopsis approved from the guide.
8. The synopsis must be brief i.e., not more that 4-6 pages. It must address details like (however, students may follow the SRS format of IEEE for writing Synopsis)
    a. Name / title of the project,
    b. Statement about the problem,
    c. Why the Particular topic is chosen? It must address Present State of the Art.

d. Objective and scope of the project,

e. Analysis, Design, Development & Testing Methodology,

f. H/W & S/W to be used,

g. Testing Technologies to be used,

h. What contribution / value addition would the project make?

i. Limitations / constraints of the project,

j. Conclusion, Future Scope for Modification,

k. References and Bibliography.

9. After approval of the Synopsis, Students will need to give Two Presentations / Demonstration, as per the schedule fixed by their respective Institutions or University. First Presentation will ideally be given after Design Phase is over. Second should be given when System Testing is over. First Presentation **(Summer Training)** should be given between _____ **and** _____. Second Presentation **(Summer Training)** is from _____ to _____**.** Final Project reports should be given latest by _____**.**

10. Summer Training report has to be submitted by _____**.**

11. The student will submit his/her project report/thesis in the prescribed format, as given hereunder, after the second presentation. The project report/thesis should include: -
    a. ONE hard copy (Maroon colour with golden print) of the project report / thesis.
    b. Soft copy of project on CD including all resource code/ compiled binary code and the manuscript in MS-Word document format.

12. Note the following guidelines with respect to Preparation of the Documentation. Please note that documentation is meant for other people, and hence it must be self-explanatory, in all respect.

12.1.1 The sequence of the Pages in the Project Report will be as follows:

1. One Transparency Sheet
2. Title Page (Strictly as per the sample supplied)
3. Institute's Certificate
4. Company's Certificate
5. Candidate Declaration of originality of work
6. Acknowledgement.
7. Abstract
8. List of Figures (Strictly as per the sample supplied)
9. List of Tables (Strictly as per the sample supplied)
10. List of Abbreviations (Strictly as per the sample supplied)

11. Contents (Strictly as per the sample supplied) and then body of the dissertation according to the content.

12.2 The pages coming under the preview of the CONTENTS will only be numbered in the BOTTOM of the Page Centrally Aligned.

12.3 ONE Hardbound Copies (One Original and Two Xerox) will be submitted with the Institute out of which one will be given back to the candidate. All the students are required to follow the same binding format in maroon color with back quote mentioning title of the project, name of the student and year.

12.4 At the end of the Project Report Two White blank sheets must be attached.

12.5 At the beginning of each chapter one blank page (Strictly as per the sample supplied) must be attached. These pages will neither be numbered nor counted in total numbering of pages. They will only indicate the beginning of a New Chapter with its learning objectives.

12.6 Font size of the documentation will be *12 Times New Roman* and the pages will be one and half line spaced. The page margin will be as under: -

Top – 1 inch, Bottom – 1 inch,

Left – 1.5 inch, Right – 1 inch.

12.7 Kindly note that all-methodological details and theoretical aspects must be written in students' own words. Copying from books or other students will not be accepted, in any case.

All students are informed not to write definition of various concepts in index, mention the topics w. r. t. to your project (i.e., how various concepts have been implemented in the project)

*** No detailed theories required**

TEMPLATE FOR CHAPTER SCHEME

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Please note that for all the System Design (Database, Input & Output) the basic Prototype, format, Table Structure, etc. is to be discussed along with related validations, verifications & normalization. However, the sample Input & Output (Screen Snapshots) will be attached in the annexure.

Under Testing, you have to discuss the approach of Testing, Test Data, Test Cases and Test Report. How Debugging has been performed, on the basis of Test Report, must be also discussed?

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CHAPTER 2

SYSTEM ANALYSIS

2.1 Feasibility Study

2.1.1 Technical Feasibility

2.1.2 Economical Feasibility

2.1.3 Operational Feasibility

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LIST OF ABBREVIATIONS

Abbreviation	Description
CAD	Context Analysis Diagram
DFD	Data Flow Diagram
HIPO	Hierarchical Input Process Output

References/Bibliography:

1. Patterson D W, "Introduction to Artificial Intelligence and Expert Systems", Second Edition, 2002, Prentice Hall of India Private Ltd., New Delhi.
2. V. Rajaraman, "An Introduction to Digital Computer Design", Third Edition, 1995, Prentice Hall of India Private Ltd., New Delhi.

Note: All of the above three, will be prepared on separate pages.

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